The Cohomological Layer of Set-Theoretic Estimation: What Gluing Buys, What It Cannot, and Where the Speedups Live

An exploratory research memo on the Čech / sheaf machinery of project `thejeb`

Project `thejeb`

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Abstract

This is an *exploratory* memo, not a results paper: it mixes a survey of machine-checked Lean 4 theorems with clearly marked conjectures, proof sketches, and open directions, and the two kinds of statement are kept typographically and verbally separate throughout. The subject is the sheaf-cohomological layer of the project's mechanization of set-theoretic estimation (STE) [11]: the constraint-side sheaf, the variable-side section presheaf and its Čech gluing obstruction [2, 3], the constant- and twisted-coefficient cochain complexes, the quantitative obstruction counts, and the rectangle-cover complexity ρ. We ask two questions. First, can the cohomological view *speed up* STE — help avoid the 2^N feasible-set blow-up, decompose, glue, or certify? Second, how does it *connect* to the rest of the development — treewidth, bucket elimination, junction trees, adaptive consistency, the dynamic-frame corpus gluing, and Carlson's cryptanalytic instantiation [9]? The centerpiece of the exploration is a short, currently *unmechanized* proposition — the *support–cover gluing theorem* — for which we give a complete proof sketch: over any cover of the variable set, the Čech gluing condition for the genuine section presheaf holds *iff* the feasible set is generated by constraints whose scopes refine the cover. It subsumes two mechanized special cases, collapses several open questions in the repository's docstrings, and, we argue, relocates the entire computational difficulty of STE into *computing local sections*, with gluing itself free. From that vantage point we identify three concrete speedup routes (one essentially proven, two conjectural), explain why the naïve hope "a cohomological invariant of the cover detects hardness" is refuted by the machine-checked acyclicity of the constant-coefficient complex, and rank the mechanization targets that would buy the most.

1 Orientation: three localities, one wall

Everything in this memo is about one wall. An STE problem [11] is a family of property sets $S_i \subseteq \Xi$ and its answer object is the feasibility set $T = \bigcap_i S_i$ (`feasibilitySet`, `mem_feasibilitySet` in `Ste.Basic`). When $\Xi = \prod_{v \in V} A_v$ is a product over N variables, T lives in a space of size exponential in N, and the project has machine-checked that the wall is real: crisp property sets can realize *any* of the 2^N extensional states (`card_allExactStates`, `arbitrary_subset_as_single_property` in `Ste.DynamicFrame.FiniteSolver`). The companion note on feasible-set shrinking catalogues the answer-preserving reductions; this memo examines the specifically *cohomological* machinery and asks what it contributes to getting around the wall.

The development contains three distinct axes of locality, and it is worth fixing them at the outset because the cohomology lives on exactly one of them.

1. **The constraint axis** (index set I). The assignment $J \mapsto \mathrm{feas}(J) = \bigcap_{i \in J} S_i$ from constraint subsets to partial feasible sets sends unions to intersections (`partialFeasibilitySet_`

`iUnion`), so any cover of the constraint index recovers the global feasible set as the limit of the local ones (`feasibilitySet_eq_iInter_cover`, in `Ste.Sheaf`). *This side is a sheaf for free*: colimits of constraints go to limits of feasible sets, and no obstruction can live here. (Under the formal-concept reading of `Ste.HyperFrame`, feas is a lower polar of the satisfaction context [18] (`partialFeasibilitySet_eq_lowerPolar`), and polars always turn joins into meets.)

2. **The variable axis** (contexts $W \subseteq V$). For a constraint $T \subseteq \prod_v A_v$ and a context W, the *local sections* $\mathcal{S}_T(W) = \texttt{localSections}\,T\,W$ are the partial assignments on W that extend to a global solution of T — the image of T under coordinate restriction (`Ste.VariablePresheaf`). Restriction along $W' \subseteq W$ makes $W \mapsto \mathcal{S}_T(W)$ a genuine contravariant functor (`restrict_restrict`, `localSections_restrict`). This presheaf is *not* in general a sheaf, and the failure is precisely variable coupling: this is where all the Čech machinery of the repository lives, and where this memo spends most of its time.

3. **The corpus axis** (document sets D, in `Ste.DynamicFrame`). The feasible-world construction $D \mapsto \text{feasible}(D)$ is contravariant in the active corpus with an *exact* binary gluing law: a hypothesis is feasible on $D \cup E$ iff it is feasible on each piece and the *cross-constraint obstruction* — the set of constraints supported only in the union and violated by the hypothesis — is empty (`crossObstruction`, `mem_feasible_union_iff` in `Ste.DynamicFrame.PresheafGluing`); a global extension of a compatible pair of local worlds exists iff that obstruction vanishes, and is then unique (`globalExtension_nonempty_iff`, `globalExtension_subsingleton`, `glue`).

The slogan the mechanization supports: *the constraint side glues for free, the corpus side glues with an exact and computable obstruction term, and the variable side is where genuine cohomology-shaped failure occurs.* Sections 2 and 3 survey what is machine-checked on the variable side. Section 4 states the support–cover gluing theorem (sketch) that we believe organizes all of it. Sections 5–6 draw the algorithmic and quantitative consequences, and Section 8 ranks what to mechanize next.

2 Inventory: the machine-checked variable-side story

All statements in this section are proved in Lean, sorry-free, in the named files; we cite theorem names exactly.

2.1 The singleton cover: gluing iff rectangular

Call T *rectangular* if $T = \prod_v P_v$ for per-variable sides P_v (`IsRectangle`). Rectangular feasibility problems are the tractable fragment: the feasible set of a rectangular family is itself a rectangle (`rectangular_feasibilitySet`) of cardinality $\prod_v |\bigcap_i P_{i,v}|$ (`rectangular_encard`), representable in $\sum_v$ space. The two-bit coupling $\Delta = \{f \mid f(0) = f(1)\}$ (`diagonal`) is provably not rectangular (`diagonal_not_rectangular`), has exactly two solutions (`diagonal_encard`), and every rectangle containing it is the whole four-point space (`rectangle_superset_diagonal_encard`): the tightest per-variable view of a coupling carries zero information.

For the cover of V by singletons, the Čech data is computed exactly (`Ste.CechObstruction`). Overlaps of distinct singleton contexts are empty (`singletonCover_overlap_empty`), so a compatible family is a point of the box $\prod_v \pi_v(T)$ (`compatibleFamilies`); a family glues iff, read as an assignment, it lies in T (`glues_iff_mem`, `gluedFamilies_eq`). The obstruction number

$$\text{obs}(T) \;=\; |\text{compatible}| - |\text{glued}|$$

(in Lean, `cechObstruction`) vanishes on rectangles (`rectangular_cechObstruction`) and equals 2 on the diagonal (`diagonal_cechObstruction`), with the mixed family ($0 \mapsto \mathsf{false}$, $1 \mapsto \mathsf{true}$) the explicit stuck cocycle (`diagonal_mixed_compatible_not_glues`). Crucially the singleton-cover story is closed as an *iff* (`cechVanishes_iff_rectangular`, in `Ste.CouplingLowerBound`):

$$\text{every compatible family glues} \iff T \text{ is a rectangle.}$$

2.2 Arbitrary covers

`Ste.CechCover` lifts the gluing level to an arbitrary cover $U : J \to \mathcal{P}(V)$ with possibly nonempty overlaps, where the compatibility condition (agreement of local sections on overlaps) is substantive (`CompatibleFamily`, `GluesCover`, `CechVanishesCover`). The headline is a genuine sheaf theorem: for a rectangular constraint and *any* cover, every compatible family of local sections glues, uniquely (`rectangular_cechVanishesCover`, `rectangular_glueCover_existsUnique`; uniqueness of glues along any cover is `glueCover_unique` and needs no constraint membership at all). Vanishing is genuinely cover-dependent: the trivial one-context cover glues everything (`cechVanishesCover_univ`), the singleton cover is the finest and hardest case and recovers the closed theory above (`cechVanishesCover_singleton_iff`), and the diagonal witnesses real failure (`diagonal_not_cechVanishesCover`). The cover-level count `coverCechObstruction` vanishes whenever the sheaf condition holds (`coverCechObstruction_eq_zero`).

2.3 Scaling the witness: the n-fold coupling

The generalized diagonal allEqual(n, α) — all n coordinates equal over an alphabet α (`allEqual`) — has every margin full (`contextSections_allEqual`), so *all* $|\alpha|^n$ families are compatible (`compatibleFamilies_allEqual_encard`) while only the $|\alpha|$ constant vectors are feasible (`allEqual_encard`). The obstruction is exactly $|\alpha|^n - |\alpha|$ (`cechObstruction_allEqual`), hence at least $2^n - 2$ once $|\alpha| \geq 2$ (`cechObstruction_allEqual_ge`): exponentially many phantom assignments that any margins-only procedure must entertain, while the true feasible set keeps constant size. For $n \geq 2$ the constraint is genuinely non-rectangular (`allEqual_not_rectangular`, `allEqual_not_cechVanishes`).

3 What the coefficients see

The Abramsky–Brandenburger program grades gluing failure cohomologically by linearizing the section presheaf [2, 3]. The repository mechanizes two coefficient regimes, and the contrast between them is, in our view, the single most instructive machine-checked fact in this layer.

3.1 Constant coefficients see nothing — a machine-checked no-free-lunch

`Ste.CechComplex` builds the constant-coefficient cochain complex on the full nerve of a cover index — modules $C^0 = \iota \to M$, $C^1 = \iota \to \iota \to M$, $\ldots$ with the standard alternating coboundaries (`cechD0`, `cechD1`, `cechD2`), the complex identities (`cechComplex_d_comp_d`, `cechComplex_d2_comp_d1`), and honest quotient modules $\check{H}^1, \check{H}^2$ (`cechH1`, `cechH2`). The vanishing theorem (`cechH1_subsingleton`, `cechH2_subsingleton`) then proves $\check{H}^1 = \check{H}^2 = 0$ over any nonempty index type: the full nerve is a simplex, and a simplex is acyclic; $\check{H}^0$ is exactly the coefficients (`cechH0EquivCoeff`).

The computational moral deserves to be stated bluntly. *Any invariant computed from the combinatorics of the cover alone — ignoring the section data — is useless for detecting coupling*:

the machine checks that it is identically zero. There is no free topological lunch; whatever a "cohomological speedup" means, its input must include the tables of local sections, i.e. exactly the data the CSP algorithms of Section 5 already manipulate. This is consistent with the contextuality literature, where the interesting classes live in relative cohomology with *presheaf* coefficients [3, 10].

3.2 Twisted coefficients see the coupling — one degree lower than expected

`Ste.TwistedCech` builds the honest twisted complex: coefficients in the linearized section presheaf $F_R(W) = (\mathcal{S}_T(W)) \to_f R$ (free R-module on genuine local sections, `twistedCoeff`), restriction by pushforward along section restriction (`sectionRes`, `twistedRes`, functorial by `twistedRes_twistedRes`), coboundaries `twistedD0`, `twistedD1` with $d^1 \circ d^0 = 0$ (`twisted_d1_comp_d0`), and $\check{H}^0$ as `twistedH0`. Two machine-checked results locate the STE coupling precisely.

1. *The AMB per-section obstruction degenerates in STE* (`amb_extension_always`). Abramsky–Mansfield–Barbosa attach to a single local section s_1 a class $\gamma(s_1) \in \check{H}^1$ vanishing iff s_1 extends to a compatible family of the linearized presheaf [3, Prop. 4.2]. The mechanization proves that in STE this extension condition holds for *every* section of *every* constraint over *every* cover — because `localSections` only ever contains restrictions of global solutions, the extension witness is free. The $\check{H}^1$-graded per-section obstruction is structurally blind here: STE's section presheaf bakes global extendability into its stalks. (The long exact sequence and the literal connecting map γ are *not* mechanized; what is proven is the extension condition AMB prove equivalent to $\gamma = 0$.)

2. *The obstruction that fires lives in the cokernel of* $F_R(V) \to \check{H}^0(U, F_R)$ (`mixedFamily_not_linearlyGlues`, `diagonal_mixed_cocycle_not_in_globalRange`). A compatible family's point-mass cochain is a twisted 0-cocycle (`familyCochain_mem_twistedH0`); the family *glues R-linearly* (`LinearlyGlues`) iff that cocycle lifts to a formal R-combination of global solutions (`linearlyGlues_iff_mem_range`). For the two-bit coupling and its singleton cover, the mixed family fails to glue linearly over *every nontrivial commutative ring* — negative and rational coefficients included. This is strictly stronger than the set-level failure (`diagonal_gluing_fails`): in the AMB world every no-signalling model has a global section over the signed reals [2], but the STE coupling is not even signed-linearly explainable by global data. Set-level gluing always implies linear gluing (`GluesCover.linearlyGlues`), and rectangles glue linearly over every cover (`rectangular_linearlyGlues`), so nonmembership is a genuine obstruction invariant (`diagonal_twisted_obstruction_detects`).

Whether the twisted $\check{H}^1$ (carried as the submodule pair `twistedCoboundaries1` $\leq$ ker `twistedD1`, with triviality expressed as `TwistedH1Trivial`; the literal quotient type is blocked by a typeclass diamond, documented in the file) vanishes for pairwise-disjoint covers is stated as *open* in the source and remains open here.

4 The support–cover gluing theorem (sketch, not yet mechanized)

Here is the centerpiece of the exploration. It occurred to us while trying to answer the prompt "does vanishing $\check{H}^1$ over a good cover give an algorithm?" and we believe it substantially reorganizes the layer. **Everything in this section is a conjecture-with-proof-sketch: none of it is mechanized**, though every ingredient it uses is, and two of its special cases are theorems in the repository.

Recall $\texttt{HasSupport}\,T\,\sigma$ (from `Ste.Support`): membership in T depends only on the coordinates in σ. Say a family $(S_i)_{i\in I}$ with supports σ_i *generates* T if $T = \bigcap_i S_i$, and say the family's scopes *refine* the cover $U : J \to \mathcal{P}(V)$ if every σ_i is contained in some context U_j.

Proposition 1 (support–cover gluing; sketch, not mechanized). *Let $T \subseteq \prod_v A_v$ and let $U : J \to \mathcal{P}(V)$ be a cover of the variable set ($\forall v\, \exists j,\ v \in U_j$).*

(a) If T is generated by a family whose scopes refine U, then $\texttt{CechVanishesCover}\,T\,U$: every compatible family of genuine local sections glues to a global solution.

(b) Conversely, if $\texttt{CechVanishesCover}\,T\,U$, then $T = \bigcap_j \pi_{U_j}^{-1}\big(\mathcal{S}_T(U_j)\big)$: T is generated by the cylinder constraints on its own local sections, a family whose scopes are exactly the contexts of U.

Hence the Čech obstruction of the cover U vanishes iff *T admits a generating family whose scopes refine U.*

Proof sketch. (a) Let (s_j) be a compatible family, each $s_j \in \mathcal{S}_T(U_j)$ with global witness $f_j \in T$, i.e. $f_j|_{U_j} = s_j$, and with s_j, s_k agreeing on $U_j \cap U_k$. Choose for each v a context $j(v) \ni v$ and define $f(v) := s_{j(v)}(v)$. Compatibility makes $f(v) = s_j(v)$ for *every* j with $v \in U_j$; in particular $f|_{U_j} = s_j$ for all j. Now take any generator S_i with $\sigma_i \subseteq U_{j_0}$. For $v \in \sigma_i$ we have $f(v) = s_{j_0}(v) = f_{j_0}(v)$, so f agrees with f_{j_0} on σ_i; by `HasSupport`, $f \in S_i$ since $f_{j_0} \in S_i$. As i was arbitrary, $f \in T$, and it restricts to the prescribed sections. (Uniqueness of the glue is already mechanized: `glueCover_unique`.)

(b) Each cylinder $\pi_{U_j}^{-1}(\mathcal{S}_T(U_j))$ has support U_j and contains T. Conversely let f lie in the intersection; then $s_j := f|_{U_j} \in \mathcal{S}_T(U_j)$ for every j, and the family (s_j) is automatically compatible (all restrict the same f). Vanishing glues it to some $g \in T$ with $g|_{U_j} = f|_{U_j}$ for all j; since U covers V, $g = f$ (again `glueCover_unique`), so $f \in T$. □

Remark 1 (Sanity anchors, all machine-checked). *Direction (b) at U = singletons is exactly* `cechVanishes_iff_rectangular`*: rectangularity = generation by singleton-scope constraints. Direction (a) contains* `rectangular_cechVanishesCover` *as the special case where the generators have singleton scopes, which refine* every *cover. The diagonal's failure at the singleton cover is consistent: its minimal scope is $\{0, 1\}$ (*`diagonal_support_full`*, in* `Ste.Support`*), contained in no singleton. And the trivial-cover success* `cechVanishesCover_univ` *is the case where the single context contains all scopes. The proposition also discharges, at the cover level, the "residual conjecture" left explicitly open in the docstring of* `Ste.VariablePresheaf` *(gluing holds precisely for the tractable fragments) — provided "tractable fragment" is read as* scope-refinement*, which per Observation 1 below is emphatically* not *the same as computational tractability.*

Remark 2 (Why no acyclicity hypothesis?). *Classical local-to-global results in constraint processing (tree-clustering, Vorob'ev-style theorems [15, 17]) require the cover's nerve to be acyclic (running intersection). Proposition 1 needs nothing of the sort, and the reason is the same structural fact behind* `amb_extension_always`*: STE's local sections are* globally extendable by definition*. The classical hypotheses are needed when the local data is merely* locally consistent *(arc- or k-consistent tables) rather than genuinely extendable; see Section 5. In Abramsky–Brandenburger terms: empirical models are arbitrary compatible presheaf data and can be contextual; the presheaf of genuine sections of a global set cannot be, in the per-section sense — but family-level gluing still fails exactly when the cover is too fine for the constraint arity.*

Corollary 1 (Cohomological scope lower bound; sketch)**.** *If some compatible family over U is stuck (nonvanishing `coverCechObstruction`), then* every *generating family of T contains a constraint whose scope is contained in no context of U. In particular the mechanized witnesses (`diagonal_not_cechVanishesCover`, `allEqual_not_cechVanishes`) certify, at the singleton cover, that no unary decomposition of these constraints exists — and a single small stuck family over any proposed bag cover certifies that a bag-local pipeline over those bags is unsound for T.*

Definition 1 (Gluing width; not mechanized)**.** *Define* $\mathrm{gw}(T)$ *as the least w such that `CechVanishesCover` $T\,U$ holds for some cover U all of whose contexts have at most $w+1$ variables. By Proposition 1,* $\mathrm{gw}(T)$ *is exactly (one less than) the minimal* arity *of a generating constraint family for T.*

Observation 1 (Gluing width is arity, not treewidth)**.** gw *must not be confused with treewidth, and the repository already contains the separating example.* $\mathrm{allEqual}(n, \alpha)$ *is generated by the* $\binom{n}{2}$ *(or even $n-1$ consecutive) pairwise-equality constraints — scopes of size 2, so* $\mathrm{gw} = 1$ *— while its primal graph is complete and its induced width is maximal ($n-1$; prose observation in `Ste.CouplingLowerBound`, not a Lean theorem). More brutally: graph 3-colorability is generated by edge-scoped disequality constraints, so it Čech-vanishes over the edge cover by Proposition 1(a); deciding whether $T \neq \emptyset$ is still NP-hard.* Vanishing of the gluing obstruction over the natural cover is therefore compatible with full computational hardness. *All the hardness has been pushed into computing the objects the proposition quantifies over — the genuine local sections $\mathcal{S}_T(U_j)$ — and none remains in the gluing.*

We regard Observation 1 as the honest answer to "is low $\check{H}^1$ over a good cover an algorithm?" It is half of one: *gluing is the free half; sections are the expensive half.* The next section shows that the repository's algorithmic layer is precisely a theory of when the expensive half is cheap.

5 Reading the algorithmic layer as section-presheaf computation

5.1 Bag tables are local sections

The treewidth chain (`Ste.Treewidth` through `Ste.JunctionTree`) is built on the `table` of a constraint on a scope σ: the image of T under restriction to σ. Comparing definitions, `table` σ T and `localSections` T σ are the *same construction* — both are the image of T under $f \mapsto (v : \sigma) \mapsto f\,v$; the identification lemma is a one-line `rfl`-adjacent statement that simply has not been written down. So the machine-checked junction-tree theorems are, without change, theorems about the section presheaf on the bag cover:

- *Faithfulness is the $\check{H}^0$ statement*: a scoped constraint is exactly the preimage of its bag table (`HasSupport.eq_preimage_table`), and a global assignment is feasible for the instance iff it restricts into every bag table (`mem_joinConstraint_iff_restrict`, `junctionTables_faithful`) — membership in T is decided by the family of its local-section sets over the bag cover. This is Proposition 1(b) in concrete clothing.
- *Cost*: a bag of at most $w+1$ variables over alphabets of size at most k tables in at most k^{w+1} rows (`table_encard_le_pow`); a width-w elimination order materializes at most $n \cdot k^{w+1}$ rows in total (`bucketEliminate_total_space`, `junctionTree_size_le`, linear-in-n form `junctionTree_size_linear` at the minimal duplicate-free width `junctionWidth`).

- *Section computation*: projective bucket elimination computes exactly the projection of the joint solution set onto the remaining variables at each step (`joinConstraint_projectBucketEliminate` and neighbours in `Ste.BucketConsistency`), decides feasibility (`projectBucketEliminate_decides`, `bucketEliminate_decides`), and extracts a solution backtrack-free (`exists_extension_of_mem_projectBucketEliminate`); the fused statement is `projectBucketEliminate_complete_solver` and, for adaptive consistency proper, `decreasingRun_adaptiveSolver` [12–14].

So the mechanized algorithmic layer already computes the genuine section presheaf over an elimination cover, in $O(n \cdot k^{w+1})$ at induced width w (`inducedTreewidth`, `achievesWidth_inducedTreewidth`; the two-sided bridge to Robertson–Seymour treewidth [22] is `treewidth_primalGraph_le` and `inducedTreewidth_le_treewidth_sub_one`; nonserial dynamic programming ancestry in Bertelè and Brioschi [7]).

5.2 Consistency algorithms compute pseudo-sections

Local-consistency algorithms [17, 20] compute something weaker than $\mathcal{S}_T$: tables that are *locally* consistent but not necessarily globally extendable. Call these *pseudo-sections.* In Abramsky–Brandenburger vocabulary a pseudo-section presheaf is exactly an empirical model over the cover [2]; contextuality — compatible local data with no global explanation — is the generic situation for pseudo-sections, and its cohomological detection is the AMB program [3, 4, 10]. The mechanized tractability results are precisely theorems that *pseudo-sections are genuine* under structural hypotheses:

- on a chain, arc consistency makes every domain value lie on a global solution (`arcConsistent_backtrackFree`, `arcConsistent_chain_solvable`; the file's own docstring already reads this as gluing over the edge cover, whose nerve is an acyclic path);
- on a rooted tree, directional arc consistency does the same (`treeArcConsistent_solvable`, `Ste.ConsistencyTree`);
- at induced width w, directional consistency of buckets does the same (`directionalConsistent_solvable`, `Ste.AdaptiveConsistency`), and bucket elimination *establishes* the condition while preserving solutions (`Ste.BucketConsistency`).

This gives a clean division of labour between the two theories:

	pseudo-sections (k-consistent tables)	genuine sections (`localSections`)
cheap to compute	yes (local closure)	only at bounded width
per-section obstruction	can be nonzero (contextuality)	always zero (`amb_extension_always`)
family gluing	fails generically	iff cover refines a generating arity (Prop. 1)
where hardness sits	pseudo $\neq$ genuine gap	computing $\mathcal{S}_T$

The conjectural summary: *the $\check{H}^1$-shaped content of STE is the pseudo/genuine gap, and every tractability theorem in the repository is a vanishing theorem for it over a structured cover.* Making "pseudo-section presheaf" a first-class Lean object and proving "RIP cover + directional consistency $\Rightarrow$ pseudo = genuine" as a single presheaf-level statement (subsuming the chain, tree, and adaptive cases) is mechanization target 2 in Section 8.

6 Quantitative invariants: obstruction counts, rectangle covers, and what is genuinely incomparable

`Ste.RepresentationBounds` settles — negatively, and machine-checked — the tempting slogan "obstruction size = forced representation blow-up." The rectangle-cover number $\rho(T)$ (`rectCoverNumber`): the least number of rectangles whose union is exactly T, the analogue of the nondeterministic communication-complexity cover number C^1 [6, 19], with the LP-extension-complexity lineage of Fiorini et al. [16], Yannakakis [23] explicitly flagged as outlook. Machine-checked:

- the fooling-set lower bound and the cardinality upper bound, $\text{fool}(T) \leq \rho(T) \leq |T|$ (`foolingNumber_le_rectCoverNumber`, `IsFoolingSet.encard_le_of_hasRectCover`, `rectCoverNumber_le_encard`); the mixing engine is `IsRectangle.mix_mem`;
- the bottom of the scale is cohomological: $\rho(T) = 1 \iff$ the singleton-cover obstruction vanishes (`rectCoverNumber_eq_one_iff_cechVanishes`, `one_lt_rectCoverNumber_iff`);
- on the coupling, $\rho(\text{allEqual}(n, \alpha)) = |\alpha|$ exactly (`rectCoverNumber_allEqual`, fooling set `isFoolingSet_allEqual`, `foolingNumber_allEqual`);
- *no uniform inequality holds in either direction* between the raw obstruction magnitude and ρ (`not_forall_cechObstruction_le_rectCoverNumber`, `not_forall_rectCoverNumber_le_cechObstruction`): rectangles give obs $= 0 < 1 = \rho$, and allEqual(3, Bool) gives $\rho = 2 < 6 =$ obs (`rectCoverNumber_lt_cechObstruction_allEqual`); at the minimal nonvanishing instance the two coincide (`diagonal_cechObstruction_eq_rectCoverNumber`) — a coincidence of the smallest example, not a law. The only equality that survives is the tautological box-gap reading (`cechObstruction_eq_box_sub_encard`).

Two representation grammars are in play and should be kept apart: ρ measures $\cup$-of-boxes (DNF-like) representations; the junction-tree machinery measures $\cap$-of-cylinders (CNF-over-bags) representations. The machine-checked triple $(\text{obs}, \rho, \text{width})$ on allEqual shows all three can diverge: obstruction exponential, ρ constant, width maximal.

Conjecture 1 (ρ vs. width incomparability, other direction; not mechanized)**.** *A width-*1 *chain of disequality constraints over an alphabet of size $k \geq 3$ on n variables has feasible set of size $k(k-1)^{n-1}$ and, we conjecture, $\rho = 2^{\Omega(n)}$ (a fooling-set construction along alternating assignments looks plausible). If so, ρ and treewidth are incomparable in both directions, and neither is a substitute for the other as a "compressibility" certificate: ρ answers "how many boxes," width answers "how large a bag."*

Practical reading for the STE practitioner: fooling sets are cheap-to-exhibit certificates that no small union-of-boxes representation of a shrunken feasible family exists; stuck compatible families over a proposed cover (Corollary 1) are cheap-to-exhibit certificates that no generating family respects those bags. Both certificate kinds are fully mechanized at the definition level and computed exactly on the coupling family.

7 The corpus axis, and the two-axis picture

The dynamic-frame gluing (`Ste.DynamicFrame.PresheafGluing`) deserves to be read side by side with Proposition 1, because it is the same theorem shape on a different index: for corpora D, E, a hypothesis is feasible on $D \cup E$ iff feasible on both pieces and the *straddling* constraints — support in $D \cup E$ but in neither piece — are satisfied (`mem_feasible_union_iff`, with the straddlers isolated as `crossObstruction`). That is exactly the two-context case of the support–cover analysis *with the obstruction term left explicit* instead of assumed away by scope refinement: when no constraint straddles, gluing is unconditional; when some do, the obstruction is a finite, per-candidate checkable set. Uniqueness of the glued world is a subsingleton instance (`globalExtension_subsingleton`) — the corpus axis' compatibility notion (identical ambient hypothesis) makes it a *separated* presheaf, which is why its gluing theory closes at $\check{H}^0$ with no higher obstruction.

Conjecturally (not mechanized, and at this point frankly speculative): the right home for both axes is a single bi-indexed presheaf on (corpora) $\times$ (variable contexts), contravariant in both, with the corpus direction separated and the variable direction carrying the twisted cohomology of Section 3. A Mayer–Vietoris-style organization of iterated corpus splits (the standard machinery, e.g. 8) would give divide-and-conquer over provenance with correction terms that are precisely the `crossObstruction` sets; since the corpus presheaf is separated, we expect the "spectral sequence" to degenerate immediately, which is a polite way of saying the corpus axis needs no cohomology beyond what is already mechanized — but a mechanized statement of the k-piece inclusion–exclusion form of `mem_feasible_union_iff` would still be a useful lemma for shard-parallel STE, and the variable-side analogue with an explicit obstruction term (splice two partial solutions, collect the straddling constraints) is mechanization target 3 below.

One more cross-connection worth recording: on the corpus axis the practical solvers (`Ste.DynamicFrame.FiniteSolver`, the TIDES/Federalist corpus instantiations in `Ste.TidesCorpus`, `Ste.FiniteInstance`, `Ste.TidesComputable`) work extensionally and hit the 2^N wall by design; the variable-side machinery of this memo is exactly what a second-generation solver would use to stay under it when the claim-interaction hypergraph has bounded arity and width. Carlson's cipher instantiation [9] sits at the opposite corner: there the hypothesis space (keys) is unstructured but the property sets are highly structured, and the productive reductions are class-level embeddings (`permutation_reduces_to_substitution`) rather than variable-local gluing.

8 Three speedup routes, ranked, and the mechanization queue

8.1 The three routes

Route 1 (essentially proven; assembly work remains): bag-local query answering. At induced width w over alphabet size k: compute the genuine section presheaf over the elimination cover by projective bucket elimination ($O(n \cdot k^{w+1})$ total table space, machine-checked as cited in Section 5); answer feasibility (`projectBucketEliminate_decides`), extract witnesses backtrack-free, and answer must/may/cannot pair queries whose variables co-occur in a bag directly from that bag's section table — soundness is faithfulness (`mem_joinConstraint_iff_restrict`) plus, for the gluing step, Proposition 1(a) (sketch). For pairs not co-occurring in a bag, propagate through separators; this is standard junction-tree practice [13] but the composite correctness statement ("the bag-local answer equals the global must/may answer") is *not yet a Lean theorem* — it is, however, within reach of the existing lemmas, and it is the statement that turns the 2^N-wall companion note's Section on structural compression into a query engine. Expected speedup: from 2^N extensional enumeration to $O(n \cdot k^{w+1})$ preprocessing plus $O(k^{w+1})$ per query, on bounded-width instances.

Route 2 (conjectural): the linear-gluing relaxation as a cheap refutation engine. `LinearlyGlues` is a linear-algebra condition; its failure refutes set-level gluing (contrapositive of `GluesCover.linearlyGlues`, machine-checked), and it is strong enough to catch the coupling over every nontrivial ring (`mixedFamily_not_linearlyGlues`). As stated, the linear system ranges over global solutions and is useless computationally; the conjecture is that over a bag cover one can present a *relaxed* system over per-bag pseudo-sections with consistency constraints on separators — a signed / affine analogue of the k-consistency hierarchy, cousin to Sherali–Adams-style CSP relaxations and to the contextual-fraction linear programs of Abramsky et al. [5] — whose infeasibility is a certificate of STE infeasibility computable in time polynomial in the table sizes. What vanishing of the twisted $\check{H}^1$ over structured covers would add is exactness guarantees for such relaxations; the open `TwistedH1Trivial` question for disjoint covers is the first stepping stone. Nothing here is mechanized; even the right statement is open, and cohomological-vs-contextual comparisons in the literature [1, 10, 21] caution that $\check{H}^1$-style invariants can have false negatives — indeed the mechanized `amb_extension_always` is precisely such a false-negative theorem.

Route 3 (proven ingredients; new composite): obstruction certificates before you pay for elimination. Before committing to an elimination order, test candidate covers: a single stuck compatible family over U (checkable on small witnesses; the repository computes them exactly for the coupling family) certifies via Corollary 1 that no generating family refines U, so no U-bag-local pipeline is sound; a fooling set certifies no small $\cup$-of-boxes representation (`IsFoolingSet.encard_le_of_hasRectCover`). Cheap NO-certificates steer the expensive machinery; this is the cohomology-as-diagnostics reading, and it is available today.

8.2 The mechanization queue

In descending expected value per line of Lean:

1. **Proposition 1** (support–cover gluing, both directions). Short (the sketch above is essentially the proof; every ingredient — `HasSupport`, `CompatibleFamily`, `glueCover_unique` — is already mechanized), and it closes the cover-level gluing question, subsumes two existing theorems, yields Corollary 1 and Definition 1, and is the correctness keystone of Route 1. Include the one-line `table` = `localSections` identification lemma while there.

2. **The pseudo-section presheaf** and the vanishing theorem "RIP cover + directional consistency $\Rightarrow$ pseudo-sections are genuine," unifying `arcConsistent_backtrackFree`, `treeArcConsistent_solvable`, and `directionalConsistent_solvable` as one presheaf statement. This is where the genuinely $\check{H}^1$-shaped content of STE lives (Section 5), and it would make the Abramsky–Brandenburger comparison exact rather than analogical.

3. **Variable-side cross-obstruction gluing**: the splice of two partial solutions over $W_1 \cup W_2$ with an explicit straddler-obstruction term, mirroring `mem_feasible_union_iff` on the variable axis; then its k-piece iteration along an RIP ordering. This is the Mayer–Vietoris lemma of Route 1's separator propagation.

4. **The bag-local query theorem** of Route 1 (must/may/cannot correctness from junction tables), and the quantitative lower-bound side the junction-tree file lists as outlook (a width-w-vs-$2^{\Omega(n)}$ separation, for which `cechObstruction_allEqual_ge` is the margin-cover prototype).

5. **Twisted $\check{H}^1$ for genuine covers**: resolve the quotient-type diamond noted in `Ste.TwistedCech`, settle `TwistedH1Trivial` for disjoint covers, and only then attempt the connecting map γ of Abramsky et al. [3] — with the caveat that `amb_extension_always` already proves the per-section γ is identically blind in STE, so the payoff is the *family-level* cokernel theory, not γ itself.

9 Summary

What is machine-checked: the constraint side is a sheaf (`feasibilitySet_eq_iInter_cover`); the variable-side section presheaf is a functor whose singleton-cover gluing characterizes rectangularity exactly (`cechVanishes_iff_rectangular`), whose arbitrary-cover gluing holds for rectangles with unique glues (`rectangular_glueCover_existsUnique`) and fails at couplings (`diagonal_not_cechVanishesCover`) with exactly computed, exponentially growing obstruction counts (`cechObstruction_allEqual`, `cechObstruction_allEqual_ge`); constant-coefficient Čech cohomology of the full nerve is provably blind (`cechH1_subsingleton`) while the twisted degree-0 cokernel class fires on the coupling over every nontrivial ring (`mixedFamily_not_linearlyGlues`) even though the AMB per-section $\check{H}^1$ obstruction provably cannot (`amb_extension_always`); the rectangle-cover number is sandwiched, characterized at 1 cohomologically, computed on the coupling, and provably incomparable to the obstruction magnitude (`not_forall_cechObstruction_le_rectCoverNumber` and its mirror); and the entire bounded-width algorithmic stack — tables, bucket elimination, junction trees, adaptive consistency — is mechanized with faithfulness and $O(n \cdot k^{w+1})$ size and a fused complete-solver statement (`projectBucketEliminate_complete_solver`).

What we conjecture: that gluing of genuine sections is governed entirely by scope refinement (Proposition 1, proof sketched, unmechanized); that in consequence the cohomological content of STE hardness is the pseudo-section/genuine-section gap, with every mechanized tractability theorem a vanishing theorem for it; that a bag-local linear relaxation with exactness controlled by twisted cohomology exists; and that ρ and width are incomparable in both directions. The single highest-value next theorem is Proposition 1: it is small, it is load-bearing for the only speedup route that is fully within reach of the current development, and — pleasingly for a memo about cohomology — its content is that in set-theoretic estimation the cohomology, properly aimed, is exactly the bookkeeping of who is allowed to talk to whom.

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